DEPI GRADUATION PROJECT DOCUMENTATION

Manufacturing Line Productivity Analysis

1. Project Planning & Management | 2. Literature Review | 3. Requirements Gathering

Working draft based on the manufacturing productivity workbook and the DEPI documentation guideline.

1. Project Planning & Management

1.1 Project Proposal

Project Title

Manufacturing Line Productivity and Downtime Analysis

Project Overview

This project analyzes the supplied manufacturing records to understand production-batch elapsed durations and recorded downtime patterns. The workbook contains production batch information, product reference data, downtime-factor descriptions, and downtime durations associated with batches. The team will prepare the data, build a suitable analytical model, define supported KPIs, and develop interactive Power BI dashboards.

Problem Statement

Production and downtime information can be difficult to analyze consistently when it is spread across multiple worksheets and downtime is stored in a matrix-like layout. The project will organize these records and provide a clearer view of recorded downtime by factor, product, batch, date, and operator where the data supports those comparisons. The available workbook alone does not establish actual production output, defect levels, financial losses, or the full impact of downtime.

Objectives

- Inspect, clean, and standardize the four source worksheets.
- Transform downtime data into an analysis-friendly structure and connect it to production batches and downtime-factor descriptions.
- Calculate valid batch elapsed durations and recorded-downtime indicators after confirming units and assumptions.
- Compare recorded downtime across factors and explore differences by product, date, batch, and operator where meaningful.
- Develop five coordinated analytical dashboard pages, subject to the team's agreed design.
- Present evidence-based findings and practical recommendations without making claims unsupported by the data.

Scope

In scope	Out of scope unless additional data is provided
Data profiling, cleaning, transformation, and modeling of the supplied workbook	Calculating actual production volume or throughput
Batch elapsed-duration analysis and recorded downtime analysis	Defect rate, scrap rate, or quality yield
Downtime-factor comparisons, including the source Operator Error classification	Claiming that a named operator caused an incident
Product, date, batch, and operator comparisons supported by available fields	Financial cost of downtime or lost revenue
Power BI dashboards and documentation	Full OEE calculation without required planned-time, ideal-cycle, total-count, and good-count data

Expected Deliverables

- Documented and cleaned data tables.
- A documented data model and KPI definitions.
- Five Power BI dashboard pages developed and integrated by the team.
- A final set of data-supported insights and recommendations.
- Project documentation and final presentation.

1.2 Project Plan and Timeline

Proposed eight-week schedule; replace this with the team's actual dates and DEPI deadlines.

Phase	Main tasks	Output	Proposed timing
1. Planning	Confirm scope, objectives, roles, risks, and communication approach	Project plan	Week 1
2. Requirements Gathering	Define stakeholders, user stories, use cases, functional and non-functional requirements	Requirements specification	Week 1
3. Literature Review	Review credible sources on manufacturing productivity and downtime analysis	Literature review and references	Weeks 1–2
4. Data Preparation & Modeling	Profile data, clean fields, reshape downtime data, validate keys and time calculations	Prepared model and data dictionary	Weeks 2–3
5. KPI Definition	Agree KPI formulas and test them against source records	KPI catalogue	Weeks 3–4
6. Dashboard Development	Build individual pages and integrate them into one report	Five dashboard pages	Weeks 4–6
7. QA & Validation	Check calculations, filters, relationships, and edge cases	Testing log and corrected report	Week 7
8. Finalization	Complete documentation, recommendations, and presentation	Final submission package	Week 8

1.3 Team Task Assignment and Responsibilities

The project work was divided among six team members. All team members collaborated on data cleaning and validation to ensure data quality and consistency. Moddather was responsible for developing the data model, while each member worked on their assigned analysis and dashboard. Ziad also served as the team leader, coordinating the team's progress and ensuring consistency across the project

Team member	Primary responsibility	Specific Tasks / Deliverables
Member 1 — Moddather	Manufacturing Production Overview & Data Modeling	Developed the data model and relationships between tables, and created Dashboard 1 to analyze overall manufacturing production
Member 2 — Fathy	Product Performance Analysis	Created Dashboard 2 to analyze production volumes, batch processing times, product characteristics, and production efficiency.
Member 3 & 4 — Alaa & Nada	Operator Performance Analysis	Collaborated on Dashboard 3 to analyze operator-related production metrics, batch processing times, and performance across products.
Member 5 — Ziad (TL)	Downtime Analysis	Created Dashboard 4 to analyze total downtime, downtime factors, frequency of downtime events, downtime trends, and batches with the highest downtime. Coordinated team activities and monitored consistency across the project.
Member 6 — Mohamed Gamal	Production Efficiency & Optimization	Created Dashboard 5 to evaluate production efficiency, compare actual and minimum batch processing times, and identify opportunities for productivity improvement.

Shared Responsibilities: All team members participated in data cleaning and validation, collaborated to resolve data quality issues, and ensured that the analyses were based on consistent and reliable data.

1.4 Risk Assessment and Mitigation

Risk	Potential impact	Mitigation
Downtime units or recording rules are unclear	Incorrect totals and comparisons	Confirm the unit and whether events overlap before final KPI calculations.
An end time is earlier than its start time because a batch crosses midnight	Incorrect or negative elapsed duration	Build full datetimes and handle overnight batches explicitly; validate the affected batch manually.
Downtime data remains in a wide matrix layout	Difficult filtering and aggregation	Unpivot factor columns into rows and preserve batch/factor identifiers.
Missing output and quality fields	Some common manufacturing KPIs cannot be calculated	State limitations and request extra data if those KPIs are required.
Inconsistent definitions across dashboard owners	Conflicting values across pages	Use one shared model and KPI catalogue; review measures centrally.
Operator Error category is misinterpreted	Unsupported conclusions about individuals	Treat it as the source's classification of a downtime factor, not proof of individual responsibility.
Late work or integration conflicts	Delayed submission	Assign clear owners, use version control, set check-in milestones, and integrate pages early.

1.5 Proposed KPIs and Success Criteria

KPI	Calculation concept	Caution / validation needed
Recorded Downtime	Sum of validated downtime durations	Confirm unit and whether durations overlap.
Downtime by Factor	Sum downtime grouped by factor description	Requires normalized downtime records and valid factor mapping.
Downtime Share (%)	Factor downtime divided by total downtime	Define behavior when the denominator is zero and clarify filter context.
Batch Elapsed Duration	End datetime minus start datetime	Handle midnight crossing; elapsed time is not necessarily machine runtime.
Batch Count	Count of distinct valid batch IDs	Check missing and duplicate IDs.
Elapsed Duration vs. Minimum Batch Time	Compare elapsed duration with product reference value	Confirm compatible units and what the reference represents.
Dashboard QA Pass Rate	Passed agreed test cases divided by all test cases	Set test cases and acceptance threshold with the team.

Project success should be judged by traceable source data, correct and documented calculations, consistent results across all five pages, usable filters, clear definitions, and recommendations supported by the analysis. Do not claim numeric performance targets until the team agrees on and tests them.

2. Literature Review

This section summarizes established concepts relevant to the project. The references below are background sources; they have not analyzed the team's Excel workbook. The team should verify the source details and follow the instructor's required citation style before final submission.

2.1 Manufacturing Productivity and Performance Measurement

Manufacturing performance is evaluated through indicators selected to match the production system and the question being investigated. No single metric describes every aspect of performance. For this project, the available records support analysis of batch elapsed duration and recorded downtime. They do not include production counts or ideal cycle output, so actual throughput cannot be established from this workbook alone.

2.2 Downtime Categorization and Measurement

Downtime analysis is more useful when interruptions are recorded consistently and assigned to understandable categories. A structured list of downtime factors can help users compare the duration associated with each recorded category. This supports the planned transformation of the Line downtime worksheet from a matrix-like layout into a normalized table with one row per batch and downtime factor. The team must confirm duration units and whether events can overlap before interpreting summed values.

2.3 Root-Cause Analysis and Continuous Improvement

Ranking downtime categories can help prioritize which processes or events deserve further investigation. However, a dashboard that shows an association or a high recorded duration does not, by itself, prove the root cause of a problem. The project should present descriptive findings from the supplied data and frame recommendations as areas for investigation unless additional evidence confirms a cause.

2.4 Manufacturing Monitoring and Data-Driven Decisions

Manufacturing monitoring and diagnostics use operational data to make performance patterns visible and support decisions. An interactive Power BI report can serve as a descriptive monitoring tool by allowing users to explore downtime by factor and compare available batch, product, date, and operator fields. Predictive maintenance is outside the current scope because the workbook does not provide sensor streams, maintenance histories, or an appropriate time series for reliable predictive modeling.

2.5 Relevance to This Project

Literature theme	How it informs this project	Limitations in the supplied data
Select performance measures that fit the question	Prioritize batch duration and recorded downtime measures that can be supported by source fields.	No production quantity, ideal cycle time, or good-unit count.
Use consistent downtime categories	Map factor IDs to their descriptions and normalize the downtime data.	Unit and event-overlap rules need confirmation.
Prioritize investigation using measured losses	Rank downtime factors and compare patterns across relevant dimensions.	Duration alone does not prove root cause or lost output.
Use monitoring to support decisions	Provide interactive views with clear labels, filters, and definitions.	No sensor data or maintenance records for predictive analysis.

2.6 Literature Review Conclusion

The literature supports a disciplined approach: define indicators carefully, structure downtime records consistently, prioritize analysis based on measured patterns, and avoid claiming causation without supporting evidence. The current workbook is appropriate for an initial descriptive analysis of recorded downtime and batch elapsed duration. Additional production, quality, planned-time, ideal-cycle, maintenance, and cost data would be required to expand the analysis to throughput, yield, full OEE, predictive maintenance, or financial impact.

2.7 DEPI Guideline Items to Confirm

The DEPI documentation guideline lists 'Feedback & Evaluation', 'Suggested Improvements', and 'Final Grading Criteria' under the Literature Review heading. These items differ from the conventional academic meaning of a literature review. The team should ask the instructor whether the submission requires an academic literature review, those three DEPI-specific items, or both. No feedback, official grading rubric, or evaluation results have been provided here, so none are fabricated.

- Feedback & Evaluation: add actual instructor feedback and the team's response after it is received.
- Suggested Improvements: document improvements supported by review and testing.
- Final Grading Criteria: insert the official rubric if provided by DEPI or the instructor.

References

1. Vorne / OEE.com. “Implement OEE: Quick Guide for Measuring OEE.” https://www.oee.com/implementing-oee/
2. National Institute of Standards and Technology (NIST). “Monitoring, Diagnostics and Prognostics for Manufacturing Operations.” https://www.nist.gov/programs-projects/monitoring-diagnostics-and-prognostics-manufacturing-operations
3. “Optimization of machine downtime in the plastic manufacturing.” Cogent Engineering (2017). https://doi.org/10.1080/23311916.2017.1335444
4. “Most important performance evaluation methods of production lines: A comprehensive review on historical perspective and emerging trends.” Computers & Industrial Engineering, 197 (2024), 110623. https://doi.org/10.1016/j.cie.2024.110623

3. Requirements Gathering

3.1 Purpose

Requirements gathering defines the information needs of intended users and translates them into testable requirements for the data model and dashboard. The stakeholders below are proposed based on the dataset and must be validated with the instructor or an actual manufacturing stakeholder if available.

3.2 Stakeholder Analysis

Stakeholder	Likely need	Potential dashboard support
Production Manager	Understand which recorded downtime categories account for the most time and compare production-time patterns.	Overview KPIs, downtime ranking, product/date filters.
Line Supervisor	Investigate a specific batch and inspect its recorded downtime details.	Batch-level details and downtime breakdown.
Process / Industrial Engineer	Identify patterns that may warrant root-cause investigation.	Factor comparisons and product/batch breakdowns.
Dashboard Maintainer / Data Analyst	Refresh, validate, and maintain the model and measures.	Documented transformations, data dictionary, KPI catalogue.
DEPI Instructor / Evaluator	Check that the project meets the agreed scope and documentation requirements.	Traceable requirements, test cases, assumptions, and limitations.

3.3 User Stories

ID	User story	Priority
US-01	As a production manager, I want to view recorded downtime by factor so that I can identify categories with the largest recorded durations.	High
US-02	As a line supervisor, I want to filter by date, product, batch, and operator so that I can investigate selected records.	High
US-03	As a process engineer, I want to compare batch elapsed duration with the product's minimum-batch-time reference so that I can identify records for further review.	High
US-04	As a dashboard user, I want KPI definitions and units to be visible so that I can interpret the results correctly.	High
US-05	As a dashboard maintainer, I want downtime factor IDs mapped to readable descriptions so that the report is understandable.	High
US-06	As a project evaluator, I want assumptions and limitations documented so that conclusions are not overstated.	Medium

3.4 Use Cases

UC-01: Explore Downtime by Factor

Field	Description
Primary actor	Production Manager / Process Engineer
Preconditions	Downtime records are normalized, factor descriptions are mapped, and the duration unit is confirmed.
Main flow	Open the dashboard; review total recorded downtime; select a factor or apply a date/product filter; inspect the resulting breakdown.
Alternative flow	If no records match the filters, show a clear no-data state rather than misleading results.
Postcondition	The user can inspect the selected factor's recorded duration and share of the filtered total.

UC-02: Investigate a Production Batch

Field	Description
Primary actor	Line Supervisor
Preconditions	Batch details, product mapping, timestamps, and associated downtime records have been checked.
Main flow	Select a batch and review its product, operator, date, elapsed duration, and recorded downtime factors.
Alternative flow	If no downtime is recorded for a batch, indicate 'no downtime recorded'; do not assume this proves no interruption occurred.
Postcondition	The user can review the available information for the selected batch.

3.5 Functional Requirements

ID	Requirement	Acceptance check
FR-01	The solution shall connect production batches to product reference data and downtime factor descriptions using validated keys.	Relationships are checked and unmatched keys are reported.
FR-02	The solution shall display the number of recorded batches.	The count reconciles to distinct valid batch IDs after documented checks.
FR-03	The solution shall calculate batch elapsed duration from start and end datetime values.	Overnight batches are handled correctly and checked against source records.
FR-04	The solution shall display total recorded downtime and downtime grouped by factor.	Values reconcile to the normalized downtime event table.
FR-05	The solution shall show readable downtime-factor descriptions instead of relying only on numeric IDs.	All factor IDs map to descriptions or are flagged as unmatched.
FR-06	The solution shall provide relevant filters for available date, product, batch, operator, and downtime factor.	Filters update applicable visuals consistently.
FR-07	The solution shall compare elapsed duration with minimum batch-time reference only after units and meanings are confirmed.	The comparison is documented and uses compatible units.
FR-08	The solution shall provide definitions and units for the displayed KPIs.	Definitions are available in the report or its accompanying documentation.
FR-09	The solution shall clearly disclose limitations and avoid unsupported throughput, yield, financial-loss, or full-OEE calculations.	The report contains no unsupported KPI claims.

3.6 Non-functional Requirements

ID	Requirement	Verification
NFR-01 Usability	The report shall use clear page titles, consistent labels, readable units, and a consistent layout.	Review the report against an agreed usability checklist.
NFR-02 Performance	Common dashboard interactions shall meet a response-time target agreed by the team.	Agree on a target and test representative interactions; do not claim a target before testing.
NFR-03 Accuracy	Displayed KPIs shall match documented calculations and validated source records.	Manually reconcile sample batches and totals.
NFR-04 Reliability	Supported filters and refreshes shall not cause broken visuals or invalid calculations.	Test normal, blank, and edge-case records.
NFR-05 Maintainability	Transformations, relationships, measures, and assumptions shall be documented.	Have another team member follow the documentation.
NFR-06 Data integrity	Cleaning decisions shall be traceable and	Retain source files and document

	the original source data shall remain unchanged.	transformations.
NFR-07 Responsible sharing	If the report is shared outside the team, access shall follow the approved sharing method.	Check access permissions and avoid unnecessary public exposure of operator names.

3.7 Data Requirements and Constraints

- Use the batch identifier to connect production records to downtime records after validating uniqueness and completeness.
- Use the product code to connect production records to product reference data.
- Use downtime-factor ID to connect downtime event rows to factor descriptions and the source Operator Error flag.
- Reshape the downtime worksheet from a wide matrix into a long table with fields such as Batch ID, Downtime Factor ID, and Downtime Duration.
- Construct complete start and end datetimes and handle overnight batches only after confirming the source's time interpretation.
- Confirm the units for downtime and minimum batch time before calculating comparisons or ratios.
- The workbook does not provide production quantity, good-unit count, defect count, planned production time, ideal cycle time, or downtime cost. Therefore, throughput, yield, full OEE, and financial loss are not supported by the current data alone.
- Treat Operator Error as a source classification attached to a downtime factor, not proof that a particular named operator caused an event.

3.8 Assumptions and Open Questions

Question / assumption	Action before final approval
What unit is used for downtime duration?	Confirm with the dataset owner or instructor.
What unit and meaning does 'minimum batch time' represent?	Confirm before comparing it with elapsed duration.
Can downtime records overlap within a batch?	Confirm before summing durations.
Are the proposed stakeholders actual intended users?	Validate with the instructor or a manufacturing stakeholder.
What are the official dashboard-page titles and requirements?	Confirm the five-page allocation with the team.
What is the approved performance target for the dashboard?	Agree on a measurable target and test it.
Which Literature Review items does DEPI expect?	Confirm whether the academic review, DEPI-specific feedback/evaluation items, or both are required.

3.9 Requirements Approval

Treat these requirements as a draft until the team and instructor confirm the scope, KPI definitions, data assumptions, dashboard-page allocation, and acceptance tests. Record any approved changes so that all five team members work from the same version.